



Documentation Report
Of Final Project/Exam:VB.NET Application Development for Sustainable
Development

Community Resource Management System: A Tool for Transparency and Accountability

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1.1 Project Overview & UN SDG Target

The Community Resource Management System is a Windows Forms application developed in VB.NET that leverages ADO.NET for persistent data handling and Report Viewer for professional reporting. Designed as part of the Capstone Synthesis for the VB.NET Application Development course, the system addresses operational challenges faced by local communities in managing sustainable initiatives.

Unlike manual record-keeping, which often results in inefficiencies and data inconsistencies, this application provides a structured, secure, and user-friendly platform for managing resources, beneficiaries, and distribution processes. By integrating object-oriented programming principles, modular architecture, and robust error handling, the system ensures reliability and maintainability in real-world deployment scenarios.

1.2 Problem Statement

Local government units and community organizations often encounter difficulties in managing sustainable resource programs:

1. **Manual Tracking Issues** – Paper-based records lead to errors, duplication, and loss of critical data.
2. **Inefficient Processing** – Verification and reporting consume significant time, delaying aid and resource distribution.
3. **Data Integrity Risks** – Lack of validation allows duplicate entries, incorrect records, and unreliable statistics.
4. **Unstructured Reporting** – Absence of automated reporting prevents stakeholders from making informed decisions.
5. **Resource Mismanagement** – Poor monitoring leads to shortages, surpluses, or inequitable allocation of resources.

Proposed Solution

The Community Resource Management System addresses these challenges through:

- **Automated CRUD Operations** – Secure database integration via ADO.NET with parameterized queries.
- **Business Logic Modules** – Dedicated classes for resource validation, beneficiary eligibility, and consumption tracking.
- **Queue and Priority Handling** – Structured distribution order ensuring fairness and prioritization of vulnerable groups.
- **Professional Reporting** – Microsoft Report Viewer integration for generating summary and trend reports.
- **Robust Error Handling** – Comprehensive Try-Catch-Finally blocks with clear, actionable error messages.
- **Data Persistence & Security** – Configuration-based connection strings and role-based login system for controlled access.

III. Objectives

3.1 General Objective

To develop a Community Resource Management System that empowers organizations with tools to manage resources efficiently, monitor consumption, and promote sustainable practices aligned with SDG 12.

3.2 Specific Objectives

1. Implement secure user authentication with role-based permissions.
2. Enable full CRUD operations for resource tracking via ADO.NET.
3. Provide a budget setup and monitoring dashboard with real-time feedback.
4. Integrate reporting through Microsoft Report Viewer for structured insights.
5. Generate smart allocation suggestions based on usage history.
6. Deliver category-based analysis and visual trend charts.

7. Support administrators with a centralized dashboard for oversight.
8. Enforce strict input validation and robust error handling.

IV. SDG Alignment

UN SDG Alignment: SDG 12 – Responsible Consumption and Production

This project directly contributes to UN Sustainable Development Goal 12: Responsible Consumption and Production, which emphasizes reducing waste, improving resource efficiency, and promoting sustainable practices. By enabling communities to track resource allocation, monitor consumption trends, and generate actionable reports, the system supports equitable distribution and minimizes mismanagement.

ID	Requirements	Description
FR1	Data Persistence	Full CRUD operations via ADO.NET with parameterized
FR2	Business Logic	Rule-based modules for allocation and monitoring.

V. Scope and Limitations

5.1 Scope

The system is a VB.NET Windows Forms application integrated with Microsoft SQL Server via ADO.NET, designed for community organizations and administrators. It supports resource tracking, budget monitoring, reporting, and role-based access.

5.2 Limitations

- No real-time API integration; data is manually entered.
- Desktop-only application; no web or mobile version.
- Rule-based suggestions only; no AI or machine learning.
- Limited to internal database; no external feeds.

VI. REQUIREMENT & ANALYSIS

6.1 Functional Requirements

ID	Requirements	Description
FR1	Data Persistence	Full CRUD operations via ADO.NET with parameterized
FR2	Business Logic	Rule-based modules for allocation and monitoring.
FR3	Reporting	Professional reports via Report Viewer.
FR4	Security	Role-based login system.
FR5	Resource Tracking	CRUD for resource entries.
FR6	Budget Monitoring	Real-time dashboard with indicators.
FR7	Smart Suggestions	Rule-based recommendations.
FR8	Admin Dashboard	Centralized oversight of system activity.
FR9	Validation & Error Handling	Strict input validation and Try-Catch-Finally

		blocks.
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6.2 Non-Functional Requirements

ID	Requirements	Description
NFR1	UI Consistency	Standard naming conventions and clean layout.
NFR2	Maintainability	XML comments and modular code.
NFR3	Reliability	Config-based connection strings.
NFR4	Performance	Handle 100+ records without lag.
NFR5	Usability	Intuitive interface for non-technical users.
NFR6	Security	Parameterized queries and password hashing.

VII. Core Features

Feature	Description	Primary User
Login & User Management	Secure authentication and role-based access	User, Admin
Resource Tracking	CRUD operations for resource entries	User
Budget Setup	Configure daily/weekly/monthly budgets	User
Budget Monitoring	Real-time comparison	User

Dashboard	of budget vs. usage	
Smart Suggestions	Rule-based recommendations	User
Category Analysis	Charts showing spending/resource distribution	User
Reporting System	Printable reports via Report Viewer	User, Admin

VIII. Tools and Technologies

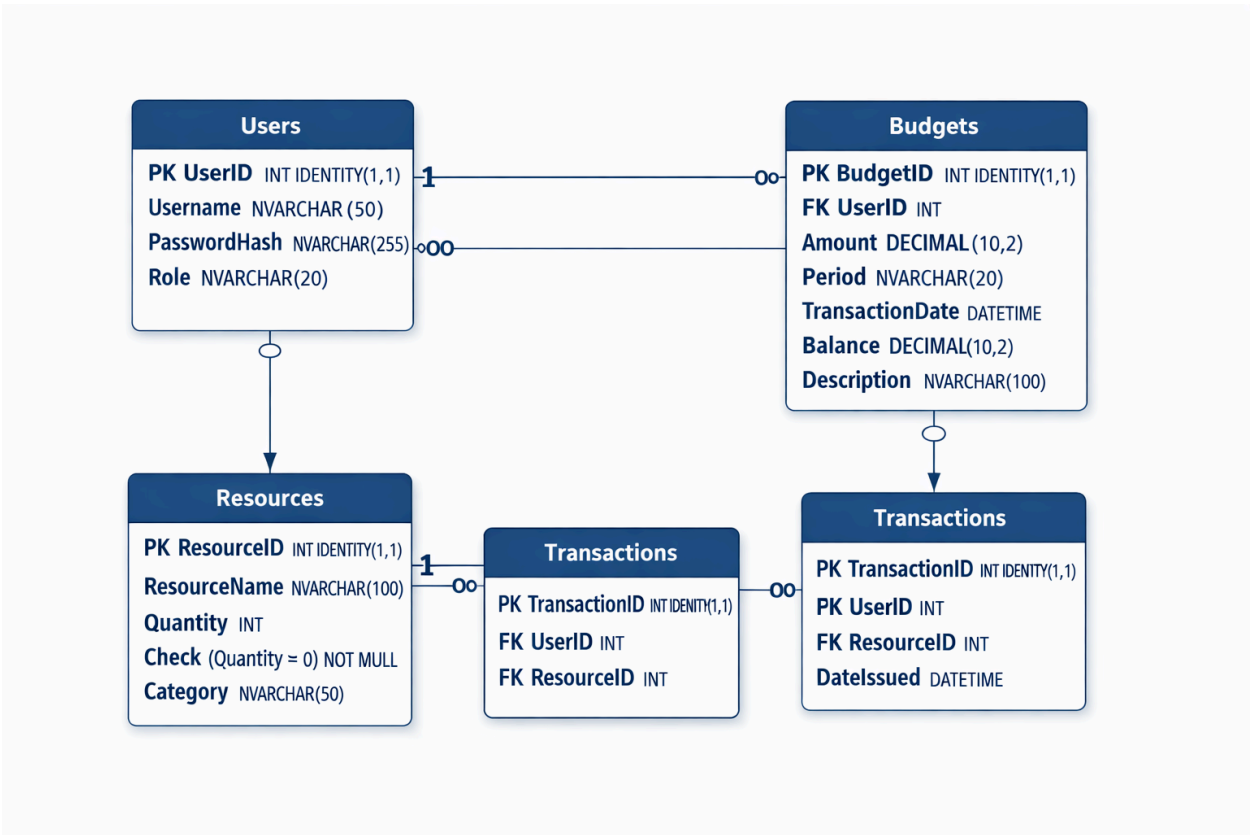
Tool / Technology	Category	Role in the System
VB.NET	Programming Language	Core application logic and forms
Visual Studio	IDE	Development and debugging
Windows Forms	UI Framework	Graphical interface
SQL Server	Database	Persistent storage
ADO.NET (SqlClient)	Data Access Layer	CRUD operations
Report Viewer	Reporting Tool	Professional reports
GitHub	Version Control	Repository and collaboration

IX. Database Design

9.1 Entity Relationship Diagram (ERD).

The Community Resource Management of ERD illustrates the logical structure of the database and the relationships among its entities. The **Users** table stores login credentials and roles, distinguishing between administrators and students. The **Resources** table contains details of available items, quantities, and categories,

supporting efficient tracking and allocation. The **Budgets** table records financial allocations per user, including amount, period, and balance, enabling monitoring of consumption and sustainability goals. The **Transactions** table logs every resource issuance and usage event, linking users and resources through foreign keys. These relationships establish a one-to-many connection between users and budgets, and between users and transactions, while resources maintain a one-to-many relationship with transactions. The ERD ensures data normalization, minimizes redundancy, and supports secure CRUD operations through ADO.NET.



9.2 Data Dictionary (Table names, fields, types, and constraints).

	A	B	C
1	Table Name: Users		
2	Description: Stores user login credentials and role classifications.		
3	Field Name	Data Type	Constraints / Description
4	UserID	INT IDENTITY(1,1)	Primary Key
5	Username	NVARCHAR(50)	Unique, Not Null
6	PasswordHash	NVARCHAR(255)	Not Null
7	Role	NVARCHAR(20)	Check (Admin, Student), Not Null
8			

1	Table Name: Resources		
2	<i>Description: Contains information about available community resources.</i>		
3	Field Name	Data Type	Constraints / Description
4	ResourceID	INT IDENTITY(1,1)	Primary Key
5	ResourceName	NVARCHAR(100)	Not Null
6	Quantity	INT	Check (Quantity ≥ 0), Not Null
7	Category	NVARCHAR(50)	Not Null
8			

1	Table Name: Budgets		
2	<i>Description: Tracks budget allocations, periods, and remaining balances for each user.</i>		
3	Field Name	Data Type	Constraints / Description
4	BudgetID	INT IDENTITY(1,1)	Primary Key
5	UserID	INT	Foreign Key → Users(UserID)
6	Amount	DECIMAL(10,2)	Not Null
7	Period	NVARCHAR(20)	Check (Daily, Weekly, Monthly), Not Null
8	TransactionDate	DATETIME	Default GETDATE()
9	Balance	DECIMAL(10,2)	Nullable
10	Description	NVARCHAR(100)	Nullable
11			

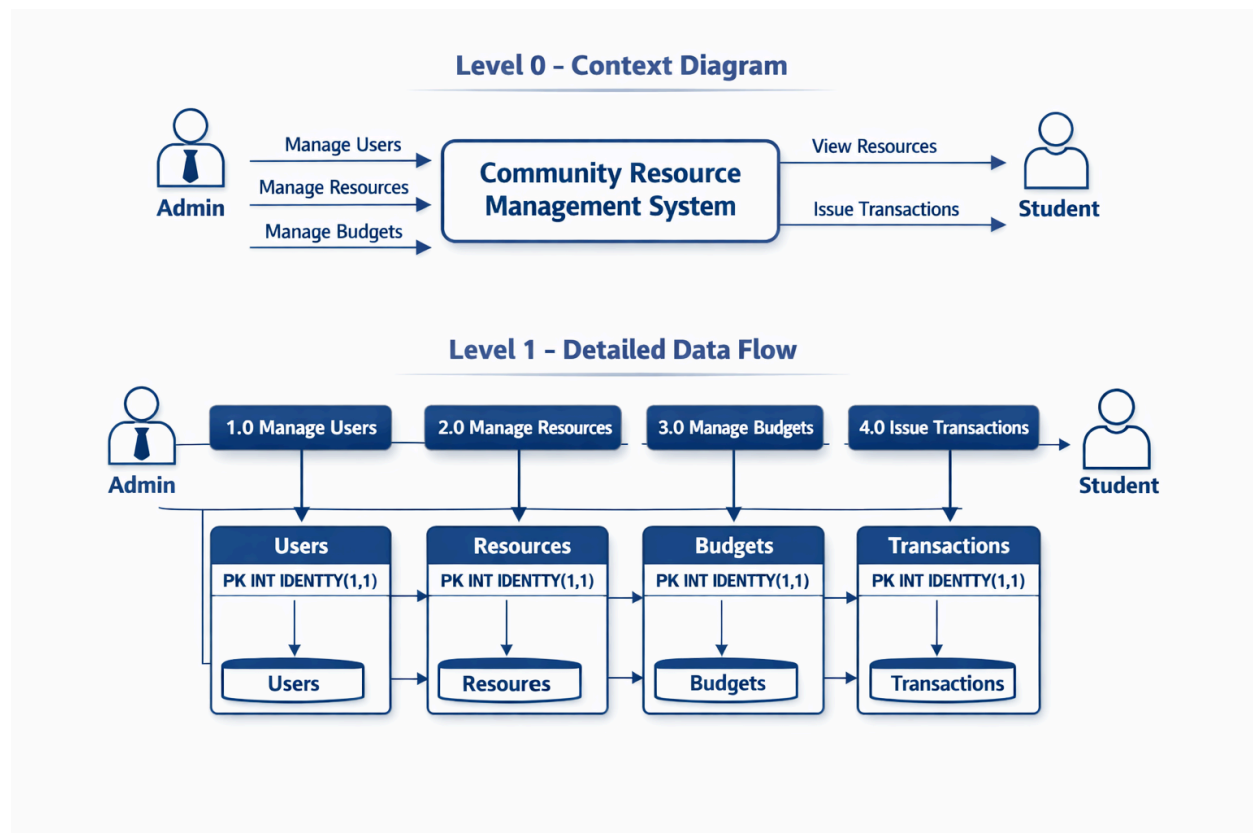
	A	B	C
1	Table Name: Transactions		
2	<i>Description: Records all resource issuance and usage events.</i>		
3	Field Name	Data Type	Constraints / Description
4	TransactionID	INT IDENTITY(1,1)	Primary Key
5	UserID	INT	Foreign Key → Users(UserID)
6	ResourceID	INT	Foreign Key → Resources(ResourceID)
7	DateIssued	DATETIME	Default GETDATE()
8			

X. System Architecture (Flowcharts showing the Data Flow (DFD Level 0 and Level 1).

The Community Resource Management System is designed using a three-tier architecture that ensures modularity, scalability, and secure data handling. The presentation layer consists of a Windows Forms interface developed in VB.NET, providing an intuitive environment for both administrators and students. It manages user interactions, input validation, and visual feedback through dashboards and forms. The

application layer serves as the core logic of the system, handling resource validation, budget computation, and transaction processing through object-oriented programming and ADO.NET integration. Finally, the data layer uses Microsoft SQL Server to store persistent data, enforcing referential integrity through primary and foreign keys. This layered architecture promotes separation of concerns, allowing each component to be maintained or upgraded independently while ensuring reliability and performance.

DFD Level 0 and Level 1 (Context Diagram) / (Detailed Data Flow)



At Level 0, the system is represented as a single process called *Community Resource Management System*, which interacts with two external entities: the Admin and the Student. The Admin manages users, resources, and budgets, while the Student views resources and issues transactions. Data flows from the Admin to the system through operations such as *Manage Users*, *Manage Resources*, and *Manage Budgets*, and from the system to the Student through *View Resources* and *Issue Transactions*. This diagram provides a high-level overview of how external actors exchange information with the system, defining clear boundaries and responsibilities.

The **Level 1** diagram expands the main process into four subprocesses: *Manage Users*, *Manage Resources*, *Manage Budgets*, and *Issue Transactions*. Each process interacts with its corresponding data store—Users, Resources, Budgets, and Transactions—to perform CRUD operations and maintain data integrity. The Admin interacts primarily with the first three processes to manage system data, while the Student interacts with the fourth process to record resource usage. Data flows between these processes ensure coordination: user management influences budget allocation, resource management connects to transaction issuance, and budget monitoring supports decision-making. This detailed view demonstrates how the system’s internal components collaborate to achieve efficient resource tracking and sustainable management.

XI. Individual Contributions

Member	Contribution Rating	Specific Contributions
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Alba, Jimmy Paul	10/10	Helped with SQL code; main creator of the project PowerPoint presentation
Doton, Athan Josch	9/10	Wrote the introduction to tools and technologies
Lizardo, Gerald	10/10	Designed the database; created system architecture designs
Navarro, Aaron Christian	10/10	Main coder for VB and SQL; organized project repository on GitHub
Santos, Justine	9/10	Assisted in making the PowerPoint; suggested system structure improvements

XII. Reference

Acknowledgment of AI Assistance

- Debugging support: Identifying and resolving SQL connection issues between the RDLC report and SQL Server.
 - [Microsoft Copilot. \(2026\). AI assistance in debugging and RDLC-SQL integration. Microsoft Corporation.](#)
- Report integration guidance: Suggesting steps to properly bind RDLC datasets to SQL queries.
 - [Microsoft Copilot. \(2026\). AI assistance in debugging SQL errors. Microsoft Corporation.](#)